

MURALIDHAR REDDY GAJJALA

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SUMMARY

AI Data Engineer and Data Science graduate with 3+ years of experience in data engineering, AI/ML solutions, and analytics across healthcare and financial domains. Skilled in Python, SQL, PySpark, Databricks, AWS, Azure, and ETL pipeline development for scalable cloud-based data platforms. Experienced in building AI-powered applications, RAG solutions, machine learning models, and business intelligence dashboards. Strong knowledge of big data processing, workflow automation, data governance, and Agile development methodologies.

EDUCATION

Saint Peter's University, The Jesuit University of New Jersey
Master of Science in Data Science (CGPA- 3.95/4)

Feb 2023- Nov 2024
Jersey City, NJ, USA

Courses: Introduction to Data Science, Data Analysis & Decision Making, Data Management Systems, Python in Data Science, Data Mining, Big Data Analytics, Machine Learning, Predictive Analytics & Finance Modelling, Data Law, Ethics & Business Intelligence, Capstone: Big Data & Data Science, Marketing Analytics & Operations Research, Applied Research Experience.

WORK EXPERIENCE

AI Data Engineer, ABBOTT
AI/LLM, RAG, MLOps, Databricks, PySpark, Lakehouse, Azure.

Jan 2026- Present
California, USA

- Built and maintained batch and real-time data pipelines using Azure Databricks, PySpark, and SQL to support analytics and AI/ML workloads across business units.
- Contributed to Lakehouse architecture design and implemented Medallion data models to organize and serve data for reporting and AI applications.
- Developed RAG-based solutions using Azure OpenAI, text embeddings, and vector search to enable intelligent document retrieval and internal chatbot features.
- Supported ML model lifecycle management using Azure ML, MLflow, and Databricks including experiment tracking, model versioning, and deployment preparation.
- Created SQL-based curated datasets and built Power BI dashboards to deliver business insights to operations and analytics teams.
- Performed Spark job tuning, data quality checks, and pipeline monitoring to ensure reliability and performance of cloud data workflows.
- Followed data governance practices using Unity Catalog and collaborated with cross-functional teams including data scientists, BI developers, and engineers in Agile sprints.

Data Engineer, Fidelity Investments
Data Engineering, PySpark, Python, SQL, AWS/Azure, ETL, ML Pipelines.

Jan 2025- Dec 2025
Texas, USA

- Designed and maintained scalable data pipelines using Python, PySpark, and SQL to process high-volume financial datasets supporting analytics, predictive modeling, and AI-driven investment applications
- Built and managed cloud data platforms on AWS and Azure including data lakes, ETL workflows, and ingestion layers to power machine learning, real-time analytics, and enterprise reporting for financial services.
- Partnered with data scientists and ML teams to engineer AI-ready datasets, build feature pipelines, and deliver curated data layers optimized for model training and inference across financial risk and forecasting models.
- Implemented data quality checks, lineage tracking, and governance frameworks to ensure accurate, compliant, and audit-ready data processing aligned with financial regulatory standards.
- Automated end to end workflow orchestration using Apache Airflow and Databricks, while tuning Spark jobs and SQL queries to reduce processing time and handle large-scale financial data efficiently.

- Integrated CI/CD pipelines for data workflow deployments, improving release reliability and reducing manual intervention in production data environments.

Data Analyst, GQBAY Software

Aug 2021- Jan 2023

Data Analysis, SQL, Python, Reporting, ETL Support.

Hyderabad, IND

- Performed advanced data analysis using SQL and Python to extract insights from large financial and operational datasets supporting business decision-making.
- Built and maintained data pipelines and transformation workflows to clean, structure, and integrate data from multiple sources including REST APIs and databases.
- Developed automated reporting systems and dashboards to track key business metrics, improve visibility, and support monthly financial reporting cycles.
- Collaborated with cross-functional teams to ensure data accuracy, resolve inconsistencies, and improve overall data quality for analytics and reporting use cases.

Software Engineer (Intern), NSIC

Nov 2018- May 2019

APIs, Python/SQL, Data Validation, CI/CD.

Hyderabad, IND

- Assisted in designing and developing RESTful APIs using Python and Java to support internal business applications and data exchange workflows.
- Built and tested data validation scripts using Python and SQL to improve data accuracy and ensure reliability of reporting and downstream systems.
- Supported CI/CD processes and code integration activities by following Agile development practices and maintaining version control using Git.

PROJECTS

Forest Fire Detection via Machine Vision (ML):

Designed and developed a modified VGG16-based deep learning model for early wildfire detection, achieving 95.5% accuracy with strong recall and real-time deployment potential on surveillance or edge systems.

Predictive Analytics and Financial Modelling:

Developed an AI-driven fraud detection system using Machine Learning techniques for data preprocessing, feature engineering, predictive modeling, and anomaly detection to identify fraudulent financial transactions with improved accuracy and scalability.

Electric Vehicle Predictive Modeling using AI:

Built an AI-powered predictive classification model using Machine Learning algorithms to analyze integrated Electric Vehicle datasets and accurately categorize EV types based on multiple technical and market features.

TECHNICAL SKILLS

Programming Languages: JAVA, C++, C, Python, JavaScript, HTML5, CSS.

Databases: MySQL, PostgreSQL, MongoDB, Elastic Search, DynamoDB, Cassandra, Redis.

Testing, Logging and Monitoring: JUNIT, Mockito, Postman, Grafana, Prometheus, Splunk, ELK stack.

Version Control and Collaboration: Git, GitHub, GitHub Actions, Gitlab, Jenkins, Agile Methodologies, Scrum, Jira, Confluence.

Technologies, Tools and Frameworks: React JS, Typescript, REST API, GraphQL, Spring, Spring Boot, Hibernate, Maven, Kubernetes, Docker, Terraform, AWS, Shell Scripting, Airflow, Spark, Linux, JSON, XML, Kafka, Microservices, Design Patterns.

CERTIFICATIONS

AWS Certified Data Engineer | Data Analysis with Python | Python 101 for Data Science | Machine Learning with Python | Game Playing AI with Shift for TensorFlow (S4TF) | Deep Learning Fundamentals | PHP Web App on a LAMP Stack | How to Build Chatbots.